

How is science different from other ways of explaining how the world works? First, science deals only with the natural world. Scientific research never concerns, in any way, supernatural phenomena of any kind. Second, scientists collect and organize information in an orderly way, looking at events for patterns and connections of cause and effect. Third, scientists propose explanations based on evidence and understanding, not belief. Some ways that scientists study the natural world are shown in **Figure 1-1**.

The Goals of Science Science is based on the view that the physical universe is composed of interacting parts and processes. From a scientific perspective, all objects in the universe, and all interactions among those objects, are governed by universal natural laws. The same natural laws apply whether the objects and events are large (like a hurricane) or small (like the cells in your body).

Greek philosophers were among the first to try to explain the natural world in terms of events and processes they could observe. Modern scientists continue that tradition. **One goal of science is to provide natural and testable explanations for events in the natural world. Science also aims to use explanations supported by data to understand patterns in nature, and to make useful predictions about natural events.**



VIDEO

Discover how scientists use scientific processes to discover the wide diversity of insects that live in people's homes.



INTERACTIVITY

CASE STUDY

Figure 1-1 Studying the Natural World

How do chimpanzees interact with one another? What is the ideal temperature for basil plants in a hydroponics system? What kinds of fish live in the Colorado River? Researchers can use science to answer these questions and many others.

